

Milestone 3: Econometric Models

QM 2023 Capstone Project

Due: Friday, Week 14 (April 24, 2026) by 11:59 PM **Points:** 50 (25% of capstone grade) **Format:** Team submission via shared GitHub repository ([main](#) branch)

Overview

Milestone 3 is the **analytical heart** of your capstone project. You transition from descriptive EDA (M2) to **causal inference** using panel regression techniques. Your goal is to estimate the causal effects of driver variables (policy shocks, market conditions, economic factors) on your outcome variable.

Real-world context: Decision-makers demand rigorous empirical evidence. A simple correlation ("[outcome] falls when [driver] rises") is not enough. You must:

- Control for confounding variables (fixed effects)
- Test assumptions (homoskedasticity, multicollinearity)
- Demonstrate robustness (alternative specifications, sensitivity analysis)
- Defend your causal identification strategy

Success criterion: Your regression models should be **publication-ready** and defensible under critical questioning.

Dataset Adaptation Note

Alternative Dataset Teams: The specifications below use REIT examples, but the **methods** apply to all panel/time-series datasets:

- **Fixed Effects (Model A)** → Works for any panel data (Entity × Time structure)
- **Model B Options** → Choose based on your research question:
 - **DiD** if you have a policy shock + treatment/control groups
 - **ARIMA** if you have a single time series (e.g., aggregate crypto index, national home prices)
 - **ML comparison** if you want to test interpretability vs. predictive accuracy

Translation Guidance: See [Dataset-Translation-Examples.md](#) for worked examples translating REIT model specs to Crypto, Housing, and Macro datasets.

Unsure which Model B to choose? If your dataset doesn't have a natural experiment or policy shock, **Option 3 (ML comparison)** works for any dataset. It only requires your existing outcome and predictor variables — no treatment/control groups needed.

Required package: Model A uses `linearmodels` for PanelOLS. Install it with `pip install linearmodels` before starting. If you have trouble installing it, `statsmodels` OLS with dummy variables can achieve similar results (ask in office hours).

Learning Objectives

By completing M3, you will:

1. **Specify and estimate panel regression models** (Fixed Effects, Two-Way FE)
 2. **Apply causal identification strategies** (DiD, time series, or ML comparison)
 3. **Diagnose assumption violations** (heteroskedasticity, multicollinearity, autocorrelation)
 4. **Implement robustness checks** (robust SEs, alternative lags, placebo tests)
 5. **Interpret coefficients economically** (not just statistically)
 6. **Communicate results** through publication-ready regression tables
 7. **Defend methodological choices** with economic reasoning
-

Deliverables

1. Python Script: `capstone_models.py`

Requirements:

- Runs from top to bottom without errors
- Uses relative paths only
- Includes clear section headers and comments
- Estimates at least **two econometric specifications**:
 - **Model A:** Fixed Effects (REQUIRED)
 - **Model B:** One of the following (CHOOSE ONE):
 - Difference-in-Differences (DiD)
 - ARIMA time series forecast
 - Machine Learning comparison (Random Forest vs. OLS)
- Outputs publication-ready regression tables (saved to results/tables/) and diagnostic plots (saved to results/figures/). Use `TABLES_DIR` and `FIGURES_DIR` from `config_paths`.

Structure:

```
"""
QM 2023 Capstone: Milestone 3 Econometric Models
Team: [Your Team Name]
Members: [List names]
Date: [Submission date]

This script estimates panel regression models to identify causal effects of
[DRIVER VARIABLES] on [OUTCOME VARIABLE]. We estimate Fixed Effects models
and [DiD / ARIMA / ML] as alternative specifications.
"""

# Section 1: Imports and data loading
# from config_paths import FINAL_DATA_DIR, FIGURES_DIR, TABLES_DIR
# Load M1 output: pd.read_csv(FINAL_DATA_DIR / '[dataset]_analysis_panel.csv')
# Section 2: Feature engineering (lags, interactions, dummies)
# Section 3: Model A - Fixed Effects regression
# Section 4: Model B - [DiD / ARIMA / ML comparison]
```

```
# Section 5: Diagnostics (heteroskedasticity, VIF, residual plots)
# Section 6: Robustness checks (robust SEs, alternative specs, placebo tests)
# Section 7: Save regression tables and diagnostic plots
```

2. Required Models

Model A: Fixed Effects Regression (REQUIRED)

Goal: Estimate the **within-entity** effect of driver variables on outcome while controlling for time-invariant entity characteristics.

Generic Specification:

$$[\text{OUTCOME}]_{it} = \beta_0 + \beta_1 \cdot [\text{DRIVER}]_{\text{lag}_{it}} + \beta_2 \cdot [\text{CONTROL1}]_{it} + \beta_3 \cdot [\text{CONTROL2}]_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

Where:

- $[\text{OUTCOME}]_{it}$ = outcome variable for entity i in time period t
- $[\text{DRIVER}]_{\text{lag}_{it}}$ = key driver variable with optimal lag (from M2 EDA)
- $[\text{CONTROL1}]$, $[\text{CONTROL2}]$ = control variables
- α_i = Entity fixed effect (controls for time-invariant unobservables)
- δ_t = Time fixed effect (controls for aggregate time shocks)
- ε_{it} = error term

REIT Example:

$$\text{ret}_{it} = \beta_0 + \beta_1 \cdot \text{fedfunds_lag2}_{it} + \beta_2 \cdot \text{MOM}_{it} + \beta_3 \cdot \text{QLTY}_{it} + \beta_4 \cdot \text{SIZE}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

Where:

- ret_{it} = return for REIT i in month t
- $\text{fedfunds_lag2}_{it}$ = Federal Funds Rate (2-month lag, per M2 EDA)
- MOM , QLTY , SIZE = factor premiums
- α_i = REIT fixed effect
- δ_t = Time fixed effect

Crypto Example:

$$\text{return}_{it} = \beta_0 + \beta_1 \cdot \text{reg_severity_lag1}_{it} + \beta_2 \cdot \text{volume}_{it} + \beta_3 \cdot \text{mcap}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

Where:

- return_{it} = token return for token i in day t
- $\text{reg_severity_lag1}_{it}$ = regulatory sentiment index (1-day lag)
- volume_{it} = 24-hour trading volume
- mcap_{it} = market capitalization

Housing Example:

$$\text{price_growth}_{it} = \beta_0 + \beta_1 \cdot \text{mortgage_rate_lag3}_{it} + \beta_2 \cdot \text{employment_growth}_{it} + \beta_3 \cdot \text{inventory}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

Where:

- price_growth_it = median home price growth for region i in month t
- mortgage_rate_lag3_it = 30-year mortgage rate (3-month lag)
- employment_growth_it = regional employment growth
- inventory_it = months of housing supply

Implementation:

```
from linearmodels.panel import PanelOLS

# Set panel structure (use your entity_id and time_var)
data_panel = merged.set_index(['[entity_id]', '[time_var]'])

# Define outcome and predictors
y = data_panel['[outcome]']
X = data_panel[['[driver]_lag[X]', '[control1]', '[control2]', '[control3]']]

# Estimate Fixed Effects model
model_fe = PanelOLS(y, X, entity_effects=True, time_effects=True).fit(
    cov_type='clustered', cluster_entity=True
)

print(model_fe.summary)
```

Expected Output:

- Coefficient table with standard errors, t-stats, p-values
- R²: within, between, overall
- F-statistic for joint significance

Key Question to Answer:

- What is the causal effect of a [1 unit / 1% / 1 pp] increase in [DRIVER] on [OUTCOME], controlling for entity and time fixed effects?

Model B: Choose One of Three Options

Option 1: Difference-in-Differences (DiD)

When to Use: You have a policy shock or event that affects a **treatment group** but not a **control group**.

Goal: Estimate the causal effect of the shock on treated entities relative to control entities.

Generic Specification:

$$[\text{OUTCOME}]_{it} = \beta_0 + \beta_1 \cdot \text{Treated}_i + \beta_2 \cdot \text{Post}_t + \beta_3 \cdot (\text{Treated}_i \times \text{Post}_t) + \text{controls} + \varepsilon_{it}$$

Where:

- $\text{Treated}_i = 1$ if entity i is in treatment group, 0 otherwise
- $\text{Post}_t = 1$ if time t is after shock, 0 otherwise
- β_3 = DiD estimator (differential effect on treated group post-shock)

REIT Example:

$$\text{ret}_{it} = \beta_0 + \beta_1 \cdot \text{Sensitive}_i + \beta_2 \cdot \text{Post2022}_t + \beta_3 \cdot (\text{Sensitive}_i \times \text{Post2022}_t) + \text{controls} + \varepsilon_{it}$$

Where:

- $\text{Sensitive}_i = 1$ if REIT i is in rate-sensitive sector (Retail, Office)
- $\text{Post2022}_t = 1$ if month $t \geq$ January 2022 (start of rate hike cycle)
- β_3 = Differential effect on sensitive sectors post-hike

Crypto Example:

$$\text{return}_{it} = \beta_0 + \beta_1 \cdot \text{DeFi}_i + \beta_2 \cdot \text{PostCrackdown}_t + \beta_3 \cdot (\text{DeFi}_i \times \text{PostCrackdown}_t) + \text{controls} + \varepsilon_{it}$$

Where:

- $\text{DeFi}_i = 1$ if token i is DeFi protocol, 0 if centralized exchange
- $\text{PostCrackdown}_t = 1$ if day $t \geq$ SEC enforcement action date

Housing Example:

$$\text{price_growth}_{it} = \beta_0 + \beta_1 \cdot \text{Urban}_i + \beta_2 \cdot \text{PostHike}_t + \beta_3 \cdot (\text{Urban}_i \times \text{PostHike}_t) + \text{controls} + \varepsilon_{it}$$

Where:

- $\text{Urban}_i = 1$ if region i is urban core, 0 if suburban/rural
- $\text{PostHike}_t = 1$ if month $t \geq$ start of Fed rate hike cycle

Key Question to Answer:

- Did the [policy shock / event] differentially impact [treatment group] compared to [control group]?

Option 2: ARIMA Time Series Forecast

When to Use: Your dataset is a **single time series** (no entity dimension) or you want to forecast aggregate outcomes.

Goal: Forecast [OUTCOME] 6-12 periods ahead using historical patterns.

Generic Specification:

```
ARIMA(p, d, q) on [OUTCOME] time series
- Use auto_arima to select optimal (p, d, q)
- Test stationarity (ADF test)
- Forecast with 95% confidence bands
```

Guidance: Aggregate to time series if panel. Use `pmdarima.auto_arima` for order selection; `statsmodels.tsa.arima.model.ARIMA` for fit and `forecast(steps=12)`. Test stationarity (ADF); report forecast accuracy vs. naive baseline.

Crypto Example: Forecast aggregate crypto market index (Bitcoin dominance-weighted).

Housing Example: Forecast national median home price growth.

Key Question to Answer:

- Can historical patterns predict future [OUTCOME]? How accurate is the forecast compared to a naive "no change" baseline?
-

Option 3: Machine Learning Comparison (Random Forest vs. OLS)

When to Use: You want to test whether complex nonlinear models improve predictions compared to interpretable linear regression.

Goal: Compare predictive accuracy of OLS vs. Random Forest; assess interpretability trade-off.

Guidance: Train/test split. Fit OLS (`statsmodels`) and `RandomForestRegressor`. Compare R^2 and RMSE on test set. Report feature importance; discuss interpretability trade-off.

Key Question to Answer:

- Does a more complex model (Random Forest) meaningfully improve predictions compared to OLS? If so, at what cost to interpretability?
-

3. Diagnostics (REQUIRED for Model A)

You must run and report the following diagnostic tests:

A. Heteroskedasticity Test

Method: Breusch-Pagan test or visual inspection of residuals

```

from statsmodels.stats.diagnostic import het_breuschpagan

residuals = model_fe.resids
# For panel models, extract fitted values manually or use OLS as proxy
# Then run het_breuschpagan(residuals, X)

```

Interpretation: If $p < 0.05$, heteroskedasticity is present. **Solution:** Use robust standard errors (`cov_type='clustered'`).

B. Multicollinearity (VIF)

Method: Variance Inflation Factor for all predictors

```

from statsmodels.stats.outliers_influence import variance_inflation_factor

vif_data = pd.DataFrame()
vif_data["Variable"] = X.columns
vif_data["VIF"] = [variance_inflation_factor(X.values, i) for i in
range(X.shape[1])]

print(vif_data)

```

Threshold: VIF > 10 indicates problematic multicollinearity. **Solution:** Drop or combine correlated predictors.

C. Residual Plots

Residuals vs. Fitted: Should show no pattern (random scatter around 0)

```

from config_paths import FIGURES_DIR

plt.figure(figsize=(10, 6))
plt.scatter(model_fe.fitted_values, model_fe.resids, alpha=0.3)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residuals vs. Fitted Values (Fixed Effects Model)')
plt.savefig(FIGURES_DIR / 'M3_residuals_vs_fitted.png', dpi=300)
plt.show()

```

Q-Q Plot: Should follow diagonal if residuals are normally distributed

```

from scipy import stats

```

```
stats.probplot(model_fe.resids, dist="norm", plot=plt)
plt.title('Q-Q Plot: Residual Normality Check')
plt.savefig(FIGURES_DIR / 'M3_qq_plot.png', dpi=300)
plt.show()
```

4. Robustness Checks (REQUIRED)

You must conduct at least 3 of the following robustness checks:

1. Robust Standard Errors

- Already implemented with `cov_type='clustered'`
- Compare clustered SEs vs. standard SEs (show both in table)

2. Alternative Lag Structures

- Test [DRIVER] at different lags (e.g., lag 1, 2, 3 for monthly data; lag 1, 7, 30 for daily data)
- Report how coefficients change across lag specifications
- **Note:** M2 EDA identified optimal lag; verify it's robust

3. Exclude Outlier Periods

- Re-estimate model excluding crisis periods (e.g., March-May 2020 for COVID, 2008-2009 for financial crisis)
- Compare coefficients: are results driven by outlier periods?

4. Group Subsamples (If Applicable)

- Estimate Model A separately for different groups (e.g., sensitive vs. resilient sectors, large vs. small entities)
- Test if driver effects differ across groups

5. Placebo Test (for DiD only)

- Run DiD regression as if policy shock occurred in earlier period (pre-treatment)
- Should find no significant effect ($\beta_3 \approx 0$, $p > 0.10$)

5. Regression Tables (Publication-Ready)

Use `stargazer` (if available) or custom formatting:

Example Table:

=====			
	Model 1: FE	Model 2: FE	Model 3: DiD
	(Baseline)	(Robust SE)	(Policy Shock)

[DRIVER] (lag X)	-0.025*** (0.008)	-0.025** (0.010)	-0.022** (0.009)
[CONTROL1]	0.042*** (0.012)	0.042*** (0.013)	0.040*** (0.012)
[CONTROL2]	0.031** (0.014)	0.031** (0.015)	0.029** (0.014)
Treated × Post	—	—	-0.035*** (0.011)

Entity FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Clustered SE	No	Yes	Yes
N	5,432	5,432	5,432
R ² (within)	0.38	0.38	0.41

*** p<0.01, ** p<0.05, * p<0.10

Save to CSV:

```
from config_paths import TABLES_DIR
results_table.to_csv(TABLES_DIR / 'M3_regression_table.csv')
```

6. Interpretation Memo: [M3_interpretation.md](#)

Required sections:

- **Model A headline:** "[1 unit/pp] increase in [DRIVER] → [magnitude] change in [OUTCOME]" with p-value. Interpret in economic units.
- **Economic interpretation:** 2-3 causal channels (e.g., leverage, discount rate, demand).
- **Model B summary:** DiD/ARIMA/ML results; key takeaway.
- **Diagnostics:** Heteroskedasticity (Breusch-Pagan), VIF, residual plots—implications and fixes.
- **Robustness:** Clustered SEs, alternative lags, outlier exclusions.
- **Caveats:** Omitted variables, parallel trends (DiD), external validity.

See [Dataset-Translation-Examples.md](#) for REIT/Crypto/Housing interpretation examples.

7. AI Audit Appendix: [AI_AUDIT_APPENDIX.md](#)

Same requirements as M1 and M2. Document all AI use, with emphasis on:

- Econometric specification decisions
- Diagnostic test interpretations
- Robustness check suggestions

Grading Rubric (50 points)

See [rubric.md](#) for detailed breakdown. Summary:

Component	Points	Criteria
Model Specification	15	Both models estimated correctly; appropriate for research question; economically sensible
Diagnostics & Robustness	12	All required diagnostics run; robustness checks thoughtful; issues addressed
Interpretation	18	Coefficients interpreted in economic terms; magnitude assessed; caveats discussed
Presentation	5	Regression tables publication-ready; code clean; memo professional

Total: 50 points

Common Pitfalls and How to Avoid Them

Pitfall 1: Ignoring Fixed Effects

Problem: Running pooled OLS instead of FE; loses causal identification **Solution:** Always include `entity_effects=True` in PanelOLS

Pitfall 2: Not Using Clustered SEs

Problem: Standard errors assume independence; panel data violates this **Solution:** Use `cov_type='clustered', cluster_entity=True`

Pitfall 3: Misinterpreting Coefficients

Problem: "A 1% increase in [DRIVER] increases [OUTCOME] by -2.5%" **Correct:** "A 1 [unit/pp/%] increase in [DRIVER] reduces [OUTCOME] by 2.5 [units/pp/%]" **Key:** Match units in interpretation to units in data

Pitfall 4: Skipping Diagnostics

Problem: Assumes Gauss-Markov holds; reports potentially invalid SEs **Solution:** Run heteroskedasticity test, VIF, residual plots for every model

Pitfall 5: No Economic Interpretation

Problem: Memo just reports " $\beta = -0.025$, $p < 0.01$ " with no context **Solution:** Explain WHY economically (mechanisms, theory, transmission channels)

Testing Checklist (Run Before Submission)

- ☐ Script runs from scratch without errors

- ☐ Both Model A (FE) and Model B (DiD/ARIMA/ML) estimated
 - ☐ All diagnostics run (heteroskedasticity, VIF, residual plots)
 - ☐ At least 3 robustness checks conducted
 - ☐ Regression tables saved to results/tables/ (CSV or text format)
 - ☐ Diagnostic plots saved to results/figures/ (PNG)
 - ☐ M3_interpretation.md complete with economic reasoning
 - ☐ AI_AUDIT_APPENDIX.md submitted
 - ☐ Coefficients make economic sense (signs, magnitudes, units)
-

Submission Instructions

Shared Team Repository Submission

1. **Work in your team's shared private GitHub repository** and submit by committing/pushing to **main**.
2. **Add your files:**

```
git add capstone_models.py
git add M3_interpretation.md
git add AI_AUDIT_APPENDIX.md
git add results/tables/M3_*.csv
git add results/figures/M3_*.png
git commit -m "M3: Econometric Models submission"
git push origin main
```

3. **Verify:** Check GitHub repo; script/tables/figures should be visible, and your latest commit should be on **main**.

Deadline: Friday, Week 14 (Apr 24) by 11:59 PM

Resources and Support

Office Hours

- **Dr. Seagraves:** Monday & Wednesday, 3:00-5:00 PM
- **Focus for M3:** Model specification, diagnostic interpretation, robustness strategies

Example Code

- **Starter script:** [starter/capstone_models.py](#)
 - **linearmodels docs:** [Panel OLS guide](#)
-

Good luck! Rigorous econometrics separates data science from data description.

Document prepared for QM 2023: Statistics II, Spring 2026, University of Tulsa Contact: Dr. Cayman Seagraves (cayman-seagraves@utulsa.edu)